



UNIVERSITÄT
LEIPZIG

Medizinische Fakultät

Information management in clinical research

MI-Lab Exercise: ARX as an anonymization tool

Leipzig, Datum

imise.

DISCLAIMER

- The exercise is largely based on a presentation of the functionalities of ARX by (then) Dr. Fabian Prasser from the Institute for Medical Statistics and Epidemiology, Technical University of Munich
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Preparation

- Introduction and installation

Basic functionalities

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- Task 2: Demonstrating the optimal transformation and finding the second-best solution
- Task 3: Comparing identification risks before/after transformation
- Task 4: Creating your own generalization hierarchy
- Task 5: Effect of the generalization hierarchy on the distribution of individual features

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Transformation models

- Task 6: Impact of suppression
- Task 7: Effect of microaggregation

INTRODUCTION

Task

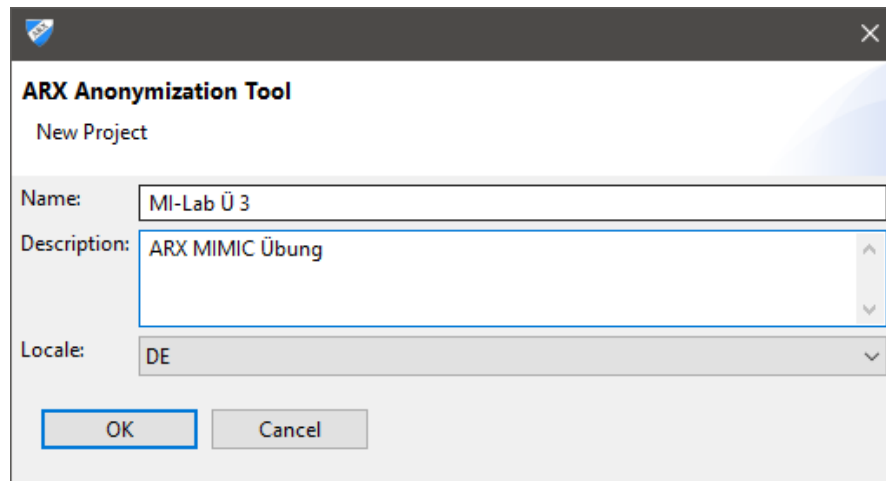
- Create an ARX project file for a given data set.

Files

- Input data
 - MIMIC Ü3 full.csv
- Hierarchies
 - Age.csv, insurance.csv, language.csv, martial-status.csv, race.csv, weight.csv

CREATE PROJCT

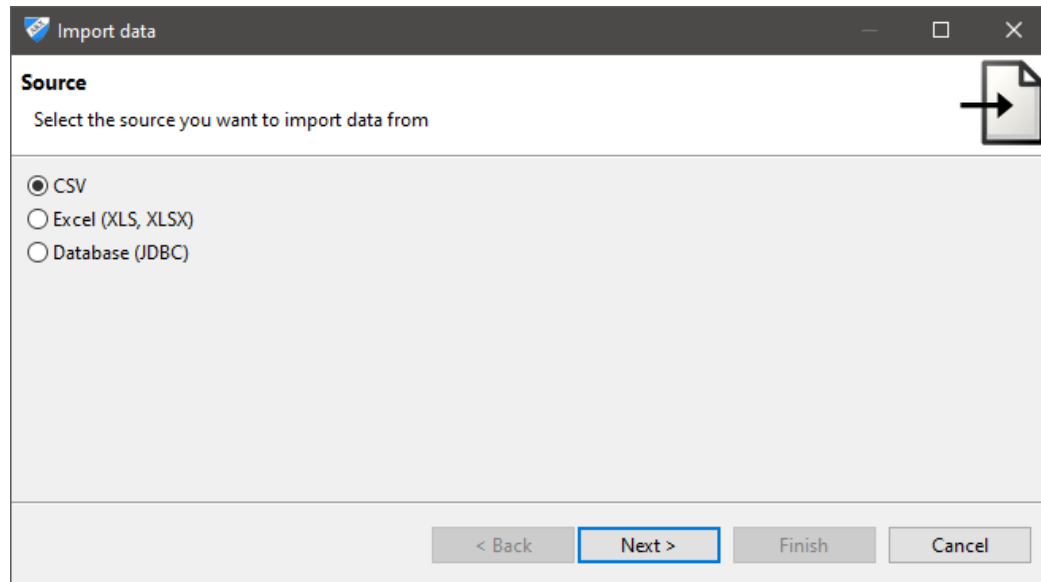
- Open ARX and select from the application menu
„File“ → „New project...“
- Enter a project name and click “OK”



The screenshot shows a dialog box titled "ARX Anonymization Tool" with a subtitle "New Project". It contains three input fields: "Name:" with the text "MI-Lab Ü 3", "Description:" with the text "ARX MIMIC Übung", and "Locale:" with the text "DE". At the bottom, there are two buttons: "OK" and "Cancel".

IMPORT DATA (1)

- Select "File" → "Import data" from the application menu.
- Select "CSV" and click "Next."
- Select "Browse..." and select the file "MIMIC Ü3.csv"



IMPORT DATA (2)

- The syntax of the file is recognized automatically.

Import data

CSV

Please provide the information requested below

Location: D:\IMISE-Arbeit\NFDI4Health\Uebung-3-ARX\MIMIC IV Demo full Ü3.csv Browse...

Charset: WINDOWS-1252 (System default)

Delimiter: ,

Quote: "

Escape: "

Linebreak: Windows

☒ First row contains column names

subject_id	age	gender	admission_type	insurance	language	marital_status	race	vor
10014729	21	F	SURGICAL SA...	Other	ENGLISH	SINGLE	WHITE - OTHE...	
10003400	72	F	EW EMER.	Medicare	ENGLISH	MARRIED	BLACK/AFRIC...	I48
10002428	80	F	OBSERVATION...	Medicare	ENGLISH	WIDOWED	WHITE	
10032725	38	F	EW EMER.	Other	ENGLISH	SINGLE	BLACK/AFRIC...	
10027445	48	F	EW EMER.	Other	ENGLISH	WIDOWED	WHITE	
10037928	78	F	OBSERVATION...	Medicare	?	WIDOWED	HISPANIC/LAT...	
10001725	46	F	EW EMER.	Other	ENGLISH	MARRIED	WHITE	
10040025	64	F	AMBULATORY...	Other	ENGLISH	DIVORCED	WHITE	I48
10008454	26	F	EW EMER.	Other	ENGLISH	SINGLE	WHITE	
10020640	91	F	EW EMER.	Other	ENGLISH	WIDOWED	WHITE	
10014078	60	F	EW EMER.	Medicaid	ENGLISH		UNABLE TO O...	

< Next > Finish Cancel

CONFIGURE INPUT FILE

- In the following dialog, you can configure the input file

Import data

Columns

Click right to edit

Selected	Name	Data type	Format
✓	subject_id	Integer	
✓	age	Integer	
✓	gender	String	
✓	admission_type	String	
✓	insurance	String	
✓	language	String	
✓	marital_status	String	
✓	race	String	
✓	vorhofflimmern	String	
✓	hyperlipidaemie	String	
✓	creatinine	String	
✓	hematocrit	Decimal	#,##0.###
✓	infusion	String	
✓	ventilation	String	
✓	ibuprofen	String	
✓	fentanyl	String	
✓	weight	String	

Move up Move down

☒ Perform data cleansing

< Back Next > Finish Cancel

PREVIEW

- An overview of the data set appears

Import data

Preview

Please check whether everything is right

subject_id	age	gender	admission_type	insurance	language	marital_status	race	vor
10014729	21	F	SURGICAL SA...	Other	ENGLISH	SINGLE	WHITE - OTHE...	
10003400	72	F	EW EMER.	Medicare	ENGLISH	MARRIED	BLACK/AFRIC...	148
10002428	80	F	OBSERVATION...	Medicare	ENGLISH	WIDOWED	WHITE	
10032725	38	F	EW EMER.	Other	ENGLISH	SINGLE	BLACK/AFRIC...	
10027445	48	F	EW EMER.	Other	ENGLISH	WIDOWED	WHITE	
10037928	78	F	OBSERVATION...	Medicare	?	WIDOWED	HISPANIC/LAT...	
10001725	46	F	EW EMER.	Other	ENGLISH	MARRIED	WHITE	
10040025	64	F	AMBULATORY...	Other	ENGLISH	DIVORCED	WHITE	148
10008454	26	F	EW EMER.	Other	ENGLISH	SINGLE	WHITE	
10020640	91	F	EW EMER.	Other	ENGLISH	WIDOWED	WHITE	
10014078	60	F	EW EMER.	Medicaid	ENGLISH		UNABLE TO O...	
10007795	53	F	DIRECT OBSER...	Medicare	ENGLISH	SINGLE	WHITE	
10001217	55	F	DIRECT EMER.	Other	?	MARRIED	WHITE	
10019172	63	F	ELECTIVE	Other	ENGLISH		UNABLE TO O...	148
10039831	57	F	SURGICAL SA...	Other	ENGLISH	MARRIED	UNABLE TO O...	
10007928	59	F	EW EMER.	Other	ENGLISH	DIVORCED	WHITE	
10019003	65	F	OBSERVATION...	Other	ENGLISH	WIDOWED	WHITE	
10002930	48	F	EU OBSERVATI...	Medicare	ENGLISH	SINGLE	BLACK/AFRIC...	
10010471	89	F	OBSERVATION...	Medicare	ENGLISH	WIDOWED	WHITE	
10027602	71	F	SURGICAL SA...	Other	ENGLISH	SINGLE	WHITE	

< >

< Back Next > Finish Cancel

ARX GUI

- Click on the attributes (columns) and change the type

The screenshot displays the ARX Anonymization Tool interface. The 'Input data' table on the left contains the following columns: subject_id, age, gender, admission_type, and insurance. The table lists 26 rows of data. The 'Data transformation' panel on the right is active, showing the 'Attribute: age' settings. The 'Type' dropdown is set to 'Quasi-identifying', and the 'Transformation' is set to 'Generalization'. The 'Minimum' and 'Maximum' dropdowns are set to 'Sensitive' and 'All' respectively. The 'Privacy models' panel at the bottom shows the 'General settings' tab, with 'Suppression limit' set to 0% and 'Precomputation' set to 'Enable'.

	subject_id	age	gender	admission_type	insurance
1	10014729	21	F	SURGICAL SAME...	Other
2	10003400	72	F	EW EMER.	Medicare
3	10002428	80	F	OBSERVATION A...	Medicare
4	10032725	38	F	EW EMER.	Other
5	10027445	48	F	EW EMER.	Other
6	10037928	78	F	OBSERVATION A...	Medicare
7	10001725	46	F	EW EMER.	Other
8	10040025	64	F	AMBULATORY O...	Other
9	10008454	26	F	EW EMER.	Other
10	10020640	91	F	EW EMER.	Other
11	10014078	60	F	EW EMER.	Medicaid
12	10007795	53	F	DIRECT OBSERV...	Medicare
13	10001217	55	F	DIRECT EMER.	Other
14	10019172	63	F	ELECTIVE	Other
15	10039831	57	F	SURGICAL SAME...	Other
16	10007928	59	F	EW EMER.	Other
17	10019003	65	F	OBSERVATION A...	Other
18	10002930	48	F	EU OBSERVATION	Medicare
19	10010471	89	F	OBSERVATION A...	Medicare
20	10027602	71	F	SURGICAL SAME...	Other
21	10031757	67	F	DIRECT EMER.	Other
22	10016742	58	F	OBSERVATION A...	Medicaid
23	10039997	67	F	SURGICAL SAME...	Medicare
24	10012853	91	F	DIRECT OBSERV...	Medicare
25	10000032	52	F	EW EMER.	Medicaid
26	10031404	82	F	URGENT	Other

IMPORTING HIERARCHIES

- „File“ → „Import hierarchy ...“ for all attributes

The screenshot displays the ARX Anonymization Tool interface. The 'Input data' table on the left contains the following columns: `subject_id`, `age`, `gender`, `admission_type`, and `insurance`. The 'Attribute metadata' panel on the right is configured for the 'age' attribute with the following settings:

- Type: Quasi-identifying
- Transformation: Generalization
- Minimum: All
- Maximum: All

The 'Level-0' through 'Level-4' table shows the generalization hierarchy for the 'age' attribute:

	Level-0	Level-1	Level-2	Level-3	Level-4
17	[15, 20[	[10, 20[	[0, 20[	*	
18	[15, 20[	[10, 20[	[0, 20[	*	
19	[15, 20[	[10, 20[	[0, 20[	*	
20	[20, 25[	[20, 30[	[20, 40[	*	
21	[20, 25[	[20, 30[	[20, 40[	*	
22	[20, 25[	[20, 30[	[20, 40[	*	
23	[20, 25[	[20, 30[	[20, 40[	*	
24	[20, 25[	[20, 30[	[20, 40[	*	
25	[25, 30[	[20, 30[	[20, 40[	*	
26	[25, 30[	[20, 30[	[20, 40[	*	

The 'Sample extraction' panel at the bottom shows the following settings:

- Size: 100 / 100 = 100%
- Selection mode: None

TASKS (SEE EXERCISE SHEET)

Exercise sheet in **GitHub**:

<https://github.com/IMISE/MI-Lab-E03-Privacy>

→ Under
/Exercise/Exercise_sheet_en.pdf





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THANK YOU!

Mona Perbix

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